



Antidiabetic effect of *Equisetum giganteum* L. extract on alloxan-diabetic rabbit

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ABSTRACT

Ethnopharmacological relevance: *Equisetum giganteum* has been traditionally used as an anti-diabetic herbal remedy to treat diabetes in the southern state of Rio Grande do Sul in Brazil.

Aim of the study: Considering the ethnopharmacology and historical importance of *E. giganteum*, its potential antidiabetic effect was evaluated in alloxan induced diabetic rabbits.

Material and methods: Samples of *Equisetum giganteum* were collected in the city of Ouro Preto, Minas Gerais, Brazil. Butanolic and aqueous extracts were prepared and subsequently evaluated for anti-diabetic properties in vivo using albino male rabbits. At the end of the treatment period, the animals were euthanized, and histopathological analysis were carried out. The following biochemical parameters were studied: glucose, triacylglycerol, cholesterol, albumin, creatinine, glycosylated hemoglobin and lipase. The phytochemical profile of the extracts was studied by liquid chromatography techniques coupled to a UV/VIS detector and high-resolution mass spectrometry.

Results: Both aqueous and butanolic extracts were capable of reducing significantly the levels of glucose, cholesterol and triacylglycerol and thus demonstrating their hypolipidemic and hypoglycemic effects. Furthermore, the extracts prevented the occurrence of hepatic complications during treatment. The phytochemical profile of the extracts was investigated, and the natural products detected were in agreement with those that had been previously described in the literature.

Conclusion: Based on the significant reductions in biochemical parameters and the histologic evidence for the absence of complications in the liver, pancreas of the treated animals, *Equisetum giganteum* can be a therapeutically relevant resource in the treatment of diabetes and hyperlipidemia.

1. Introduction

Diabetes mellitus is characterized by increased blood glucose levels (hyperglycemia) and caused by insufficient or inefficient insulin secretory response (Prabhakar, 2011). Sufferers from diabetes are estimated to reach 366 million by 2030, most of which will be from developing countries and aged between 45 and 64 years of age (Roglic, 2004). Diabetes is accompanied by severe oxidative stress, which is caused by increased oxygen free radical production (Baburao and Anand, 2012; Xiong et al., 2019) and antioxidants have been shown to be beneficial in preventing complications associated with diabetes (Zatali and Sanusi, 2013). Medicinal plants have been used for centuries

in folk medicine to treat diabetes as complementary and alternative remedies (Jin-Ming et al., 2019). Thus, they may also serve as an interesting source of potential drug candidate compounds (Pereira et al., 2011; Newman and Cragg, 2012; Caro-Ordiers et al., 2020). For this reason, plant species demonstrating anti-diabetic properties have been subjected to phytochemical studies and in return provided a large number of anti-diabetic compounds (Martins et al., 2004; Lamba et al., 2000; Bai et al., 2019). Marles and Farnsworth (Marles and Farnsworth, 1995) listed 1200 species of plants that have been used to treat diabetes worldwide. The genus *Equisetum giganteum* has been traditionally used as an anti-diabetic herbal remedy to treat diabetes in the southern Brazilian state Rio Grande do Sul. Traditional healers use the aerial part

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of the dried plant by boiling it in water and drinking the tea daily (Vendruscolo et al., 2005).

The genus *Equisetum* is cosmopolitan, colonizer of deforested areas, found on the banks of lakes and swampy environments (Ricco et al., 2011). They are widely grown for ornamental purposes and folk medicine (Raven et al., 1996). In Brazil, *Equisetum giganteum* L. is commonly called *cavalinha* but also known in other parts of the world as giant horsetail, swamp-horsetail, snake corn, fleshy grass-tail, alligator cane, sandpaper-plant amongst other aliases. *Equisetum giganteum* L., is a nonflowering erect, arborescent herbaceous perennial, with a rough texturized green hollow stem that may reach 5 m high, 2.4 cm in maximum diameter (Ricco et al., 2011; Hauke, 1990). The adult plant consists of underground rhizome, erect photosynthetic branches and well defined internodes (Joly, 1991).

Phytochemical characterization of *Equisetum giganteum* L. stems revealed a total ash content of 11.85% and a total phenolic and flavonoid content of 7.27 mg GAE/g (Francescato et al., 2011). Among the secondary metabolism products reported for the genus *Equisetum*, one can highlight the pyridine alkaloids, nicotine, palustrine, flavonoid glycosides of apigenin, quercetin and kaempferol derived from chlorogenic acid, caffeic and tartaric acid (Gruenwald et al., 2000) and the presence of thiamine (Soraru and Bandoni, 1978). *Equisetum* extracts have been studied for their hypoglycemic activity (Cetto et al., 2000; Revilla et al., 2002; Safiyeh et al., 2007), inhibition of platelet aggregation (Mekhfi et al., 2004; López-Lázaro et al., 2011), vasodilator (Sakurai et al., 2003), antioxidant activity (Ricco et al., 2011), diuretic effects (Wright et al., 2007) antimicrobial activity (Radulović and Stojanović, 2006 - Kloucek et al., 2005), antiviral activity (Churqui et al., 2018) anti-inflammatory (Jabeur et al., 2017), antiadherent activity (Alavarce et al., 2015), and incorporated into a denture adhesive against *C. albicans* biofilm (Almeida et al., 2018). Because of these pharmacological effects, studies on the chemical composition of *E. giganteum* stems were conducted (Francescato et al., 2013). The major phenolic compounds of *E. giganteum* identified by liquid chromatography coupled to diode array detection (LC-DAD) and liquid chromatography coupled to electrospray ionization-tandem mass spectrometry (LC-ESI-MS/MS) were glycosylated kaempferol derivatives. Other natural products identified include caffeic acid derivatives, flavonoids and styrylpyrones. Rarer phenolic components, namely, quercetin-3-O-(caffeoyl)-glucoside and 3-hydroxyhispidin-3,4'-di-O-glucoside were also reported in the *Equisetum* genus. The findings of these studies demonstrated that this plant has many bioactive compounds relevant to the treatment of diabetes syndrome. Polyphenols have hypoglycemic activity, inhibitory effect on the activity of the enzyme aldose reductase (responsible for the development of diabetic cataract) and on platelet aggregation (Ivorra et al., 1989; Laddha et al., 2019). Thus, this study was designed to explore the antidiabetic capacity of the *Equisetum giganteum* butanolic and aqueous extracts in alloxan induced diabetic rabbits.

2. Materials and methods

2.1. Plant material

Samples of *Equisetum giganteum* were collected in September 2004 in the city of Ouro Preto, Minas Gerais, Brazil. A voucher specimen of the plant was identified by taxonomist Dr. Alexandre Salino (Federal University of Minas Gerais) and deposited in the José Badini Herbarium at the Federal University of Ouro Preto (registry number: 16,999). The plant was dried at room temperature without exposure to sunlight and then pulverized in a mechanical mill.

2.1.1. Preparation of the aqueous and butanolic extracts

The method for preparing both the aqueous and butanolic extracts and the rationale for their concentrations was based on a procedure previously described by Cetto and co-workers (Cetto et al., 2000). Their study revealed that the extracts of *Equisetum myriochaetum* at doses of 7

and 13 mg/kg significantly reduced ($p < 0.001$) the glucose level in streptozotocin induced diabetic rats. Thus, based on studies reported in the literature (Cetto et al., 2000; Revilla et al., 2002), the preparation method of the extracts as well as the dosage for this experiment was decided: for each extract a suspension was prepared at a concentration of 17 mg/kg and 3.5 mg/kg. In order to prepare these samples, 11.9 g of each extract was needed to prepare 500 ml of suspension at a concentration of 17 mg/kg and 2.5 g of extract for the same volume at a concentration of 3.5 mg/kg. Butanol was chosen for its capacity to extract a broad range of bioactive compounds with pharmacological properties. In addition, the use of water as a solvent is justified given the fact that diabetic people use it popularly as a decoction.

In preparing the butanolic extract, 420 g of plant material was soaked in hexane (500 ml) for 24 h. Hexanes were decanted and the plant material was further treated methanol in a Soxhlet extractor for 48 h. The methanol extract was evaporated to dryness on a rotavapor under reduced pressure. Next, the green residue was dissolved in a mixture of $\text{CCl}_4/80\%$; MeOH 1:1 H_2O . The organic phase was discarded and the MeOH- H_2O phase was evaporated. The residue formed was dissolved in 1:1 mixture of butanol and H_2O , lyophilized to provide an oily residue that was stored at 4 °C. Finally, the extract was made up to a 500 ml suspension with a concentration of 3.5 mg/kg (using 2.5 g of the extract) and 17 mg/kg (using 11.9 g of the extract) body weight with the following additives: 7.5 mg of sodium carboxymethyl cellulose (CMC), followed by 2 ml of Tween, 10 ml of propylene glycol, 10 ml of a preservative (methylparaben and propylparaben) and 500 ml MilliQ water.

In obtaining the aqueous extract, 210 g of the plant material was treated with distilled water (500 ml) for 4 h under reflux, followed by filtration and evaporation of the solvent. Next, the sweet smelling brown coloured extract was dried at reduced pressure and stored at 4 °C (Cetto et al., 2000).

2.1.2. Preparation of the suspensions of the aqueous and butanolic extracts

The extract was made up to a 500 ml suspension with a concentration of 3.5 mg/Kg (using 2.5 g of the extract) and 17 mg/kg (using 11.9 g of the extract) body weight with the following additives: 7.5 mg of sodium carboxymethyl cellulose (CMC), 2 ml of Tween, 10 ml of propylene glycol, 10 ml of a preservative (methylparaben and propylparaben) and 500 ml MilliQ water. Upon addition of each reagent, the extract was homogenized. The suspensions were stored in amber glass bottles properly identified and labelled.

2.2. In vivo studies

For this study, 2-months-old male Albino rabbits (*Oryctolagus cuniculatus*) were used, breed in New Zealand and weighing between 1.285 kg and 2.055 kg (average weight of 1.68 kg). They were kept in individual cages at room temperature, free access to water and fed with 120 g/day of commercial pellet growth (Soma Ltda Food Industry and Commerce). The first five days were reserved for adaptation of the animals to the laboratory.

After 24 h of fasting, the animals received an intraperitoneal injection of a suspension of alloxan (100 mg/kg of body weight) dissolved in 0.9% of NaCl, except the control group (Oliveira et al., 2002). Alloxan induces two sequential phases before the diabetic state itself. Stage I occurs shortly after administration of alloxan, characterized by an initial hyperglycemia (adrenergic response), with decreased plasma levels of insulin and increased plasma levels of cortisol and glucagon. The second phase is characterized by hypoglycemia associated with an increase in plasma insulin levels, attributable to the release of insulin by beta cell degeneration. Five days after the induction of diabetes with alloxan, fasting glycemia (12 h) was measured and the animals with blood glucose above or equal to 180 mg/dL were considered diabetic and distributed between the groups. The body weight of the animals was recorded using a digital balance (Digipeso DP3000) on the day they

arrived at the Biopharmaceutical Laboratory and at 7, 15, 16, 21 and 23 days from the beginning of the experiment.

To avoid hypoglycemia induced death, three doses of 10 ml of 50% glucose w/v, at intervals of 4, 8 and 12 h were administered. Group 1 (normal rabbit) was administered only saline NaCl. The diabetic state was determined 48 h after application of alloxan. Blood samples were taken from the retro-orbital venous plexus with a capillary aliquot collected in a Duran tube containing 100 µL of disodium EDTA and centrifuged. Next, 500 µL of serum was separated and serum glucose level was quantified on a multiparametric autoanalyzer equipment Biochemistry - Alizé (LISABIO B652) with Bioclin kit. The animals with glucose levels above 180 mg/dL, displayed symptoms such as polyuria, polydipsia and weight loss and were therefore considered diabetic. In this study, two experiments were carried out. The first consisted of assessing antidiabetic activity of the aqueous extract at doses of 3.5 mg/kg and 17 mg/kg over a period of 23 days. In the case of the butanolic extract, doses of 3.5 mg/kg, 17 mg/kg and the control groups were evaluated over a 21-day period.

For the antidiabetic activity test with the aqueous extract the animals were divided into the following groups: **Group 1 (C)**: group of normal animals that received only food (control group); **Group 2 (D)**: group of diabetic animals without treatment; **Group 3 (DEA 3.5)**: group of diabetic animals treated with 1.2 ml of aqueous extract at a concentration 3.5 mg/kg; **Group 4 (DEA 17)**: group of diabetic animals treated orally with 1.2 ml of aqueous extract at a concentration 17 mg/kg; **Group 5 (DC)**: group of diabetic animals treated with capsule of Diabinese (chlorpropamide - Pfizer Ltd.) at a concentration of 5 mg/kg.

The antidiabetic activity test with the butanolic extract the animals were divided into the following groups: **Group 1 (C)**: group of normal animals that received only food (control group); **Group 2 (D)**: group of diabetic animals without treatment; **Group 3 (DEB 3.5)**: group of diabetic animals treated with 1.2 ml of butanolic extract at a concentration 3.5 mg/kg; **Group 4 (DEB 17)**: group of diabetic animals treated with 1.2 ml of butanolic extract at a concentration 17 mg/kg; **Group 5 (DC)**: group of diabetic animals treated with capsule of Diabinese (chlorpropamide - Pfizer Ltd.) at a concentration of 5 mg/kg.

The doses of suspensions prepared from the aqueous and butanolic extracts were administered orally, via gavage (forced feeding). By the sixteenth day of treatment (aqueous extract) or the fifteenth day of treatment (butanolic extract), the rabbits were fasted for 12 h and a blood sample was taken from the venous plexus and collected in a Duran tube containing 100 µL of disodium EDTA (anticoagulant). The material was centrifuged at 7100 × g for 15 min and 500 µL of plasma removed to allow quantification of glucose (Bioclin), triglycerides (Bioclin), cholesterol (Biolab), albumin (bioMérieux Vitek, Inc.) and creatinine (Bioclin) levels on the Alizé equipment (Lisabio B652). After 23 days of treatment (aqueous extract) or 21 days of treatment (butanolic extract), the rabbits were euthanized in a chamber containing ethyl ether. The blood was taken by cardiac puncture with 5 ml syringes, centrifuged 7100 × g for 15 min. Blood glucose levels, triglycerides, cholesterol, albumin, creatinine were quantitated after 16 and 23 days (aqueous extract) or 15 and 21 days (butanolic extract) of treatment by Biochemical multiparametric measurements Alizé. Glycosylated haemoglobin was analyzed by ion exchange resin system and spectrometer readings in the range of 410–415 nm.

2.3. Statistical analysis

Analyses were performed using the R2.3.1 (R Development Core Team 2006) program. As the experimental design consisted of different individuals and repeated measurements for each of the individuals (longitudinal data), mixed models were adjusted following the package procedure “nlme” in R. Biochemical parameters (creatinine, triglyceride, albumin and cholesterol) and the post-treatment time (16 and 23 days for the aqueous extract, 15 and 21 days for the butanolic extract), treatments were the response variables. The interactions

between them were considered fixed factors and the animal as random factor. Blood glucose concentration was also analyzed using the same model, however it was regarded as continuous variable. From a complete model, non-significant terms were removed and then the change in F was verified in a suitable minimal model, using the Likelihood-ratio test (LRT). The concentrations of hemoglobin and glycosylated lipase were compared only at 21 days via a non-parametric Kruskal-Wallis test.

2.4. Phytochemical profile

A 5 mg sample of butanolic extract was completely dissolved in MeOH. The aqueous extract was dissolved in MilliQ water and HPLC grade MeOH (8:2). In both cases a final concentration of 5 mg/ml was obtained. The samples were shaken on ultrasound (Ultra Sonic Cleaner - Unique) for 15 min and centrifuged in a microfuge (Scientec 14.000D) for 10 min at 10,000 rpm rotation. A chromatogram of the extracts was obtained by reverse phase HPLC with diode array detector. A ODS column (250 × 4.0 mm ID, 5 µM) was used (Merck, Germany) at a temperature of 40 °C. LED-UV Detection was performed at 210 nm. For the mobile phase, a flow rate of 1.0 ml/min was employed for a linear gradient composed of H₂O (A) and CH₃CN (B): 0 min 95% A, 5% B; to 5% A, 95% B over 60 min, then maintained for 5 min as isocratic elution at this ratio. Samples were dissolved in MeOH:H₂O (2:8) (for analysis of the aqueous extract) and MeOH (100% butanolic for analysis of the extract) to a concentration of 5 mg/ml. Upon centrifugation, 10 µL was injected into the HPLC.

Hydrolysis of the extracts was carried out by taking 0.5 g of either the butanolic or the aqueous extract and dissolving it in 10 ml of H₂O/MeOH solution (1:1) and 2 ml of 6 M HCl. The solutions were heated to 80 °C while stirring for 30 min. Upon completion of the reaction, the mixture was poured into a separating flask, together with 12 ml of water and 24 ml of dichloromethane. The organic phase was separated and washed with water and sodium bicarbonate solution (10%) until the pH reached 7. An aliquot of the organic solution was analyzed by mass spectrometry (HPLC-MS-IT-TOF (Shimadzu)), electrospray ionization (3.5 kV, gas nebulizer 1.5 L/min; dry gas 100 KPa) and ionization induced by interface collision at 200 °C. Analyzed *m/z* range 150–1000. Column: VP-ODS column (reverse phase), size 150 mm and 2.0 µm, ID: 3 µm, column temperature: 40 °C, flow rate: 0.2 ml/min, mobile phase composed of H₂O (A) and Methanol (B): 10% B to 70% B over 15 min then hold 70% B for 10 min. Next, 70%–100% B over 2 min then 100% B to 10% B over 3 min, injection volume: 2 µL.

2.5. Histological analysis

At the end of the treatment period, the animals were euthanized in a chamber containing Ethyl Ether (Isifar Industry and Chemicals Ltda). Their abdominal cavity was opened, the liver and pancreas identified, collected and fixed in 4% neutral buffered formalin, dehydrated, cleared, embedded in paraffin, cut into 4–5 µm sections and stained with hematoxylin and eosin (H&E) for histopathological analyses. The qualitative assessment of the organs was carried out using a microscope at 200X and 400X magnification.

3. Results and discussion

3.1. Aqueous extract

The glucose concentration in the blood plasma varied with treatment ($F = 5.464$, $p < 0.001$), with time ($F = 32.017$ $p = 0.003$) and with interaction between treatment and time ($F = 2.814$, $p = 0.034$). In all groups, there was an increase in glucose concentration in blood plasma over time (Figs. 1 and 2).

No difference was found between groups 2 and 3 (test L. ratio = 0.171; $p = 0.92$) and between groups 1 and 4 (LRT = 2.598;

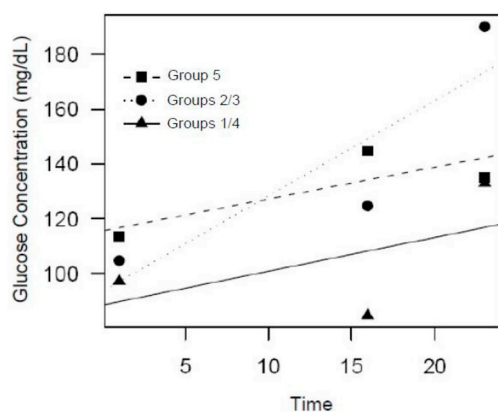


Fig. 1. Average glucose concentration over time.

$p = 0.27$). In the initial time, group 5 had a higher concentration of glucose than groups 2 and 3 and groups 1 and 4. However, over time the plasma glucose concentration in groups 2 and 3 was higher than the other groups (Fig. 1).

An analysis of variance followed by the Dunnett's test were carried out (Table 1). At 16 days the G3 displayed a glucose reduction of 29.86%, group G4 reduced glucose levels by 43.91% and at 23 days G4 reduced glucose by 31.26% when compared to G2. A similar reduction at 16 days for G3 and G4 when compared to G5 was also observed. Notably, G4 showed the greatest percentage reduction by reducing glucose 43.91% and 31.26% in comparison to G2 (Fig. 2 and Table 1).

Grover et al. (2000) mentioned that the reduction in plasma glucose is dependent on the protocols for diabetes induction that vary in the intensity of hyperglycemia (Grover et al., 2000). This fact would imply that the hypoglycemic effect of the plant extract is dependent on the dose of the diabetogenic agent and the degree of destruction of the beta cell (RAO et al., 2001). Various mechanisms of action have been described to explain how plants or their extracts act to reduce the glucose rate (Alvarez et al., 2004; Coskun et al., 2005; Marles and Farnsworth, 1995; Huo et al., 2003; Said et al., 2002; LI et al., 2004; Volpato et al., 2002) and how flavonoids exhibit hypoglycemic action (Chakravarthy et al., 1980, 1983; Ribeiro et al., 2004; Zarzuelo et al., 1996; Ong and Kloo, 2000).

In the present study, the histopathological evaluation of the pancreas demonstrated that there was no total destruction of the islets, thus, enabling the production and release of insulin in animals that were subjected to induced diabetes (Fig. 6). Therefore, a high rate of plasma glucose was not expected in these animals, which can be seen in Figs. 1 and 2.

Adrenaline, released in stress situation promotes glycogen degradation and in turn elevates the concentration of glucose in the blood (Harper, 1977). This fact would help to understand similar values among animals treated with extract at the dose of 3.5 mg/kg and without treatment (G2), as well as elevated glucose values, at 16 days, in the group treated with the drug chlorpropamide (G5) (Fig. 1). The oral hypoglycemic agent chlorpropamide increases insulin secretion by acting on membrane receptors, inhibiting potassium efflux. Depolarization of the membrane favors the influx of calcium and subsequent stimulation of insulin secretion (Cook, 1995). Thus, we can assume that the concentration of 3.5 mg/kg of the extract was not sufficient to produce a hypoglycemic effect in rabbits, possibly due to the low concentration of the crude extract, and therefore of the bioactive substances. In tests with medicinal plants, it is common to use doses above those used in our experiment, studies carried out with another species of the genus *Equisetum*, *E. arvense* to evaluate the hypoglycemic effect doses of 50, 100, 250 and 500 mg/kg were administered of the crude extract for five weeks (Safiyeh et al., 2007). In the aqueous extract of *Equisetum giganteum*, molecular-based patterns belonging to the flavonoid classes were identified (Figs. 14 and 15 Table 9), which can be inferred that the glucose reduction in animals treated with aqueous extract is based on the presence of flavonoids where the association of its various substances can exert a synergistic effect and, therefore, potentiate the hypoglycemic effect.

The concentration of triglyceride (Fig. 3) did not vary significantly between groups G1, G3 and G4 ($F = 2.09$; $p = 0.15$). Likewise, triglycerides levels for rabbits treated with chlorpropamide (G5) and the untreated diabetic rabbits (G2) did not differ from each other ($F = 1.681$; $p = 0.19$). However, G2 and G5 groups had higher plasma triacylglycerol than the other groups. In general, the triglyceride concentration was highest after 23 days of treatment ($F = 32.723$; $p < 0.001$ comparing 23 days to initial 16 days). These results indicate that the aqueous extract showed hypotriglyceridemic activity, since the animals treated with the extract were statistically similar to the control

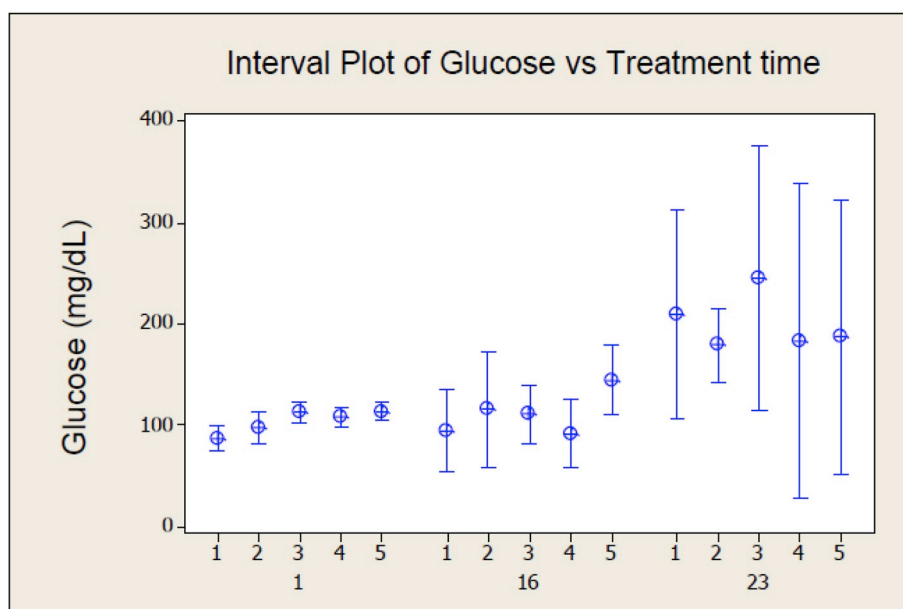


Fig. 2. Mean values and confidence intervals for the blood glucose levels the control group (G1), untreated diabetic rabbits (G2), diabetics rabbits treated with aqueous extract at a concentration of 3.5 mg/kg (G3), 17 mg/kg (G4) and 5 mg/kg Chlorpropamide (G5) at 16 and 23 days.

Table 1

Percentage of glucose variation for all groups in relation to the diabetic group without treatment and the diabetic group treated with chlorpropamide after the 16th and 23rd days of treatment.

Groups	GLUCOSE			Groups			
	1 Days	16 Days	23 Days		1 Days	16 Days	23 Days
G2				G5			
G1	−11.04%	−40.72%*	−20.67%*	G1	−23.55%*	−39.91%*	+5.43%
G3	+15.15%	−29.86%*	+11.81%	G3	−1.04%	−28.91%*	+48.6%*
G4	+10.64%	−43.91%*	−31.26%*	G4	−4.92%	−43.14%*	−8.65%
G5	+16.37%	−1.34%	−24.76%*	G2	−14.07%	+1.36%	+32.9%*

Legend: The averages with * differ from the G2 standard and the G5 standard at the 5% probability level by the Dunnet test. G1: Normal group. G2: Diabetic group without treatment. G3: Diabetic group that received aqueous extract at a concentration of 3.5 mg/kg. G4: Diabetic group that received aqueous extract at a concentration of 17 mg/kg. G5: Diabetic group treated with the drug chlorpropamide.

group (G1).

Elevated values of triacylglycerol for the untreated diabetic group were expected, since excess glucose is used for the synthesis of triacylglycerol. The G4 group showed the highest percentage of reduction for triacylglycerol (Table 2) displaying reductions of 50.42% and 33.74% at 16 and 23 days respectively when compared to the G2 group. When compared to G5, once again G4 gave the highest percentage of reduction, 50% at 16 days and 23.36% at 23 days. This result can be seen in the histological aspects of the pancreas in diabetic animals, there was a reduction in the size of the endocrine portion of the pancreas, i.e., the pancreatic islets were visibly smaller in these animals (Fig. 6).

The clinical significance of these reductions is important, since patients with diabetes also suffer hypertriglyceridemia and use drugs to treatment this for long periods, especially those in the fibrate and genfibrozil class. Thus, the fact that the aqueous extract of *E. giganteum* promotes a reduction in triacylglycerol is advantageous since usually a combination of drugs is used to reduce triacylglycerol and glucose whereas in this case only one plant extract was able to reduce these two parameters.

It has been reported that flavonoids compounds have the ability to decrease blood lipid concentration (Nagem et al., 1994, 1995). The ability of flavonoids to increase lipase activity is seen as one of the mechanisms of action to explain the triacylglycerol reductions (Oliveira, 2002). The characterization of the phytochemical profile of

Table 2

Percentage variation of triacylglycerol for all groups in relation to the diabetic group without treatment and the diabetic group treated with chlorpropamide after the 16th and 23rd day of treatment.

Groups	Triacylglycerol		Groups		
	16 Days	23 Days		16 Days	23 Days
G2			G5		
G1	−46.80% ^a	−64.57% ^a	G1	−46.35% ^a	−59.02% ^a
G3	−42.77% ^a	−27.90% ^a	G3	−42.29% ^a	−16.61%
G4	−50.42% ^a	−33.74% ^a	G4	−50% ^a	−23.36% ^a
G5	−0.84%	−13.54%	G2	+0.84%	+15.65%

^a Values differ from the G2 standard and the G5 standard at the 5% probability level (Dunnet test). G1: Normal group. G2: Diabetic group without treatment. G3: Diabetic group that received aqueous extract at a concentration of 3.5 mg/kg. G4: Diabetic group that received aqueous extract at a concentration of 17 mg/kg. G5: Diabetic group treated with chlorpropamide.

the aqueous extract revealed through the HPLC-UV and HPLC-MS the presence of the flavone, flavonol and flavanone classes (Figs. 15 and 16 and Table 9). The action of these chemical constituents suggests that the reduction in the level of triacylglycerol in rabbits treated with aqueous extract of *E. giganteum* is due to their presence.

Diabetic induced rabbits (G2) had higher concentrations of cholesterol (Fig. 4) in comparison to other treated groups ($F = 7.077$;

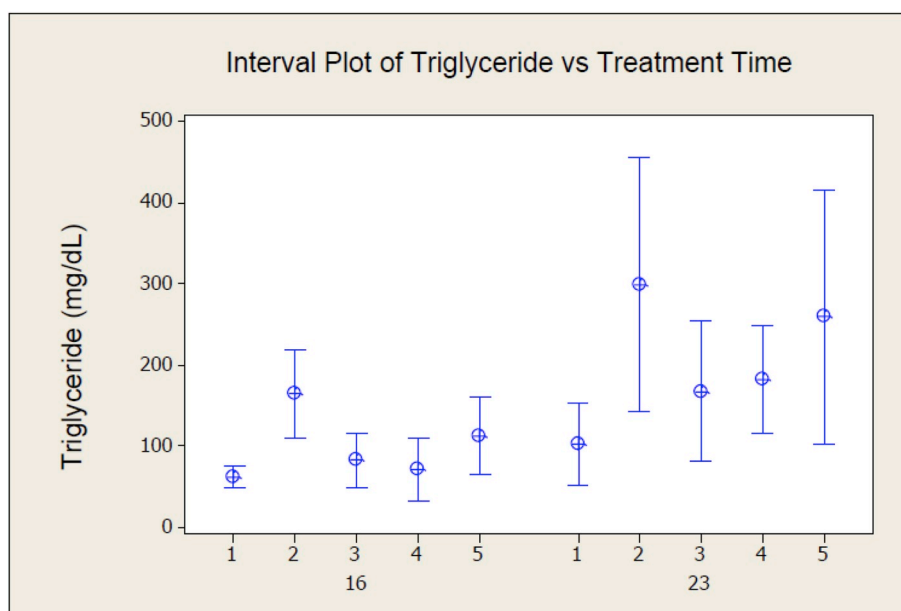


Fig. 3. Mean values and confidence intervals for the blood triglyceride levels the control group (G1), untreated diabetic rabbits (G2), diabetics rabbits treated with aqueous extract at a concentration of 3.5 mg/kg (G3), 17 mg/kg (G4) and 5 mg/kg Chlorpropamide (G5) at 16 and 23 days.

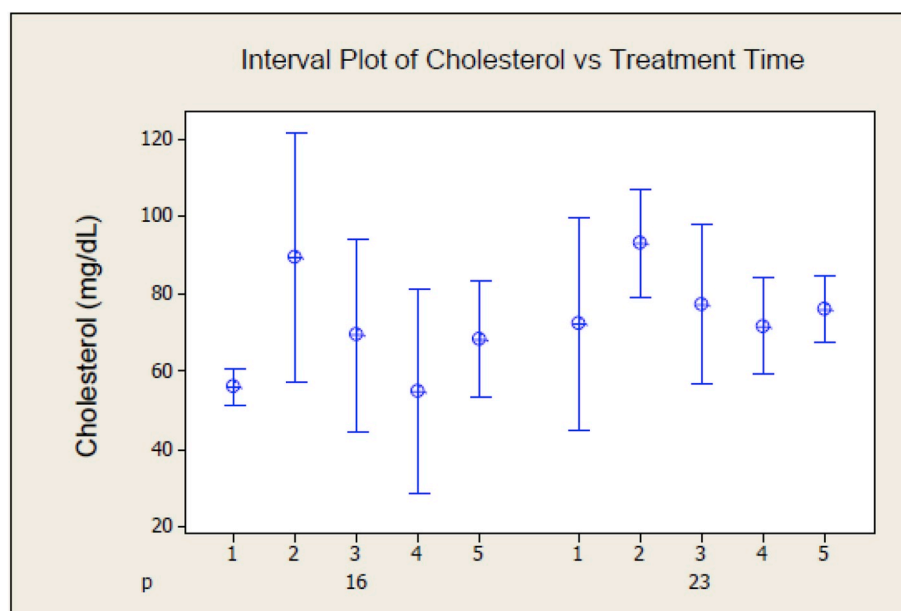


Fig. 4. Mean values and confidence intervals for the blood cholesterol levels the control group (G1), untreated diabetic rabbits (G2), diabetics rabbits treated with aqueous extract at a concentration of 3.5 mg/kg (G3), 17 mg/kg (G4) and 5 mg/kg Chlorpropamide (G5) at 16 and 23 days.

$p = 0.02$). A difference in cholesterol not was observed between the groups treated with the aqueous extract (G3 and G4) and the control group (G1) ($F = 0.21$; $p = 0.88$ and $F = 0.13$, $p = 0.90$, respectively). Cholesterol levels were highest after 23 days and the difference from 16 days was calculated ($F = 6.026$; $p = 0.02$).

The cholesterol variation in relation to the G2 group and in relation to the G5 group are presented in Table 3. In animals treated at a concentration of 17 mg/kg, G4 showed the highest reduction values when compared to the treated diabetic groups (G3 and G5) and the untreated diabetic group (G2). When comparing each group with the untreated diabetic group (G2), it was observed that G4 reduced cholesterol by 38.65% at 16 days and 19.10% at 23 days.

Hyperglycemia and diabetic hypercholesterolemia are processes responsible for high morbidity and mortality in the diabetes pathology (Sebbag et al., 1994; Jin-Ming et al., 2019). The increase in levels of triacylglycerol and cholesterol is one of the main consequences of insulin resistance (Roche, 1999). The control of cholesterol synthesis is exercised over certain enzymes at the beginning of the catabolic pathway. By controlling the activity of HMG-CoA, cholesterol synthesis can be reduced by different mechanisms: feedback inhibition, hormonal regulation and inhibition by drugs and herbal medicines (Ludke and

López, 1999 and Lopes et al., 2000a,b; Bai et al., 2019). Many polysaccharides have been identified as bioactive components responsible for hypoglycemic activity and hypocholesterolemic effect (Yuan et al., 1998; Wang & NG., 1999). The presence of polysaccharides in plant extracts may be one of the causes of the hypolipidemic effect. Flavonoids, such as quercetin, inhibit LDL oxidation and cytotoxicity *in vitro* (Lackeman et al., 1986). Glycosides of quercetin and rutin showed an inhibitory effect on lipid peroxidation (Takahama, 1984; Hwang et al., 2017), while quercetin, reduced lipid levels (Kato & Tosa, 1983). Isoflavonoids also showed a reduction in lipid peroxidation of fatty acids and inhibition in the formation of malonaldehydes (Nishiyama et al., 1993). Quercetin, dihydroquercetin, 3-methylquercetin rutin, toxirutin and fisetin are potent inhibitors of platelet action in animals. Therefore, chemical constituents such as flavonoids and polysaccharides, also present in the aqueous extract of *E. giganteum*, act in the prevention of late complications of diabetes. In addition, lipids (cholesterol and triacylglycerol) when compared to the blood glucose values, it can be suggested that the extract is more effective in combating lipid dysfunctions, such as hypercholesterolemia and hypertriglyceridemia (Figs. 2–4).

Serum levels of albumin ($F = 89.768$, $p < 0.001$) and creatinine ($F = 12.073$, $P = 0.02$) were greater after 23 days but no difference was noted between treatments ($F = 2.000$ albumin; $p = 0.13$, creatinine $F = 1.093$, $p = 0.38$). Moreover, no significant interaction between treatment and time on serological albumin ($F = 2.411$; $p = 0.07$) and creatinine concentration ($F = 1.558$, $p = 0.22$) was observed. Upon completing 23 days of treatment, no difference in the serological level of glycosylated hemoglobin of the rabbits was observed. These results, according to the experiments, may indicate that there were no renal changes in diabetic animals (treated or not) that compromised glomerular filtration. In addition, the results suggest that liver cells did not undergo significant changes that would compromise the synthesis of albumin, as was also observed in the hepatic morphology (Fig. 5). The chronic lack of control of plasma glucose can result in chronic renal failure. While creatinine may increase due to renal failure, both urea and creatinine are used to detect manifestations of kidney damage (Ragan, 1989; Lima, 2004). Our results from albumin and creatinine demonstrated that the control animals were no different from diabetics, indicating that the duration of diabetes did not cause damage to the kidneys, although their values have increased over time.

Table 3

Percentage of variation of cholesterol for all groups in relation to the diabetic group without treatment and diabetic treated with chlorpropamide after the 16th and 23rd days of treatment.

Groups	CHOLESTEROL		Groups		
	16 Days	23 Days		16 Days	23 Days
G2			G5		
G1	−37.29%*	−22.39%*	G1	−17.85%*	−4.73%
G3	−22.60%*	−8.16%	G3	+1.41%	+12.73%
G4	−38.65%*	−19.1%*	G4	−19.62%*	−0.69%
G5	−23.67%*	−22.82%*	G2	+31.01%*	+22.75%*

Legend: The averages with * differ from the G2 standard and the G5 standard at the 5% probability level by the Dunnet test. G1: Normal group. G2: Diabetic group without treatment. G3: Diabetic group that received aqueous extract at a concentration of 3.5 mg/kg. G4: Diabetic group that received aqueous extract at a concentration of 17 mg/kg. G5: Diabetic group treated with the drug chlorpropamide.

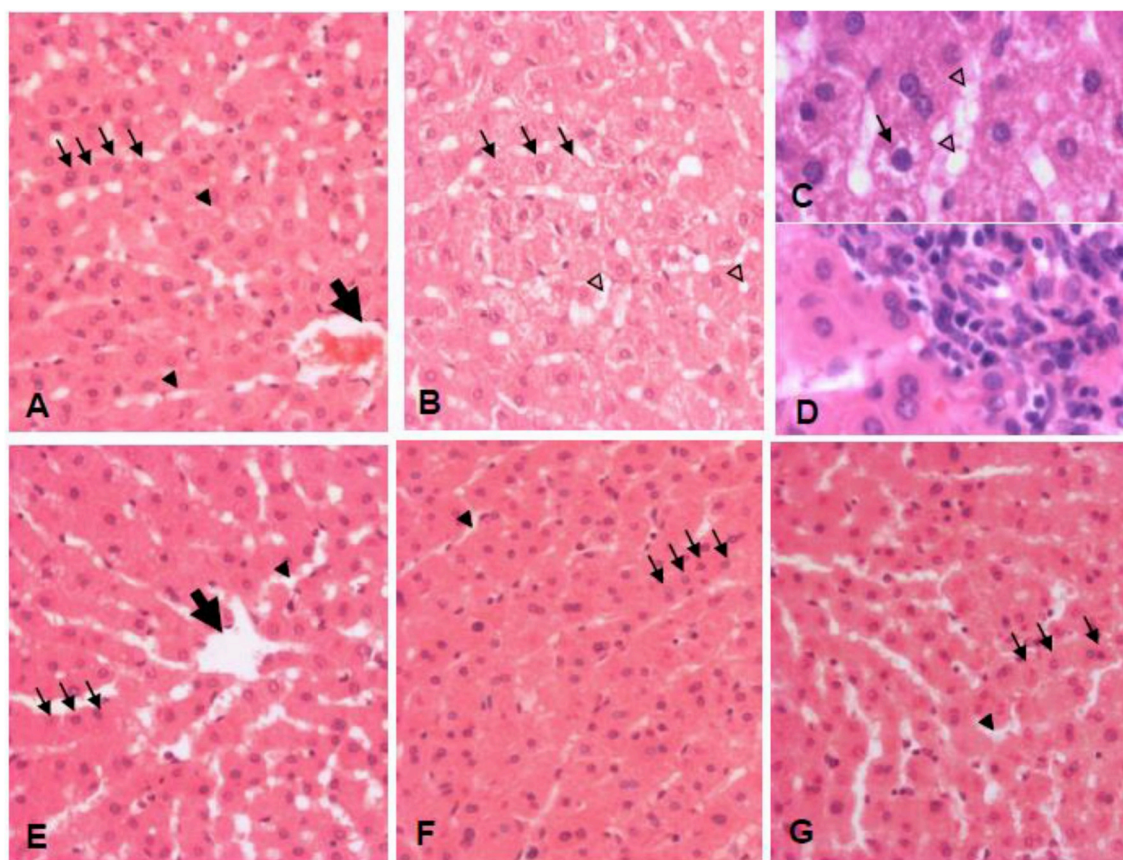


Fig. 5. Histopathological aspects of liver. Photomicrographs of liver sections stained with hematoxylin and eosin. 200X magnification (A, B, E, F, G) and 400X magnification (B, C). Representative image of the liver of control rabbits (A), untreated diabetics (B, C, D) and diabetics treated with 3.5 mg/kg of aqueous extract (E), 17 mg/kg of aqueous extract (F) and with 5 mg/kg of chlorpropamide (G) after 23 days of treatment. **A:** Control liver (G1) presents preserved liver parenchyma showing central vein (thick arrow) and hepatocyte plaques (thin arrows) limiting spaces occupied by well-preserved sinusoid capillaries (arrowheads). **B:** Alloxane-diabetic liver without treatment (G2) showing hepatocyte plaques (thin arrows) containing enlarged cells with a swelling aspect (steatosis), more frequent in these animals when compared to the other treatments (thin arrow). Sinusoidal dilatation (empty arrowheads). **C:** Details of alloxane-diabetic liver without treatment (G2) that presents large hepatocytes with morphological aspect of steatosis (arrow) and sinusoidal dilatation (empty arrowheads). **D:** Detail of the periportal region of the alloxane-diabetic liver without treatment (G2), containing an inflammatory process with a predominance of mononuclear cells (right region). **E, F and G:** Alloxane-diabetic liver treated with 3.5 mg/kg of G3 extract (E), 17 mg/kg of G4 extract (F) and 5 mg/kg of chlorpropamide G5 (G) with similar histological characteristics those in the control group. Central vein (thick arrow), hepatocyte plaques (thin arrows), well-preserved sinusoid capillaries (arrowheads).

The comparative histology in the liver between the groups showed a similar pattern in the animals of the G1, G3, G4 and G5, indicating the histological characteristics of a normal liver (Fig. 5). In the G2 group, hepatocytes presented enlarged, with morphological aspect of steatosis and were observed hepatic sinusoidal dilatation.

Histological analysis of the control pancreas (G1), alloxane-diabetic pancreas without treatment (G2) or treated with aqueous extract at a concentration of 3.5 mg/kg (G3) and 17 mg/kg (G4) is shown in Fig. 6. There was no histological difference between untreated and treated diabetic animals. Histopathological evaluation revealed that the diabetic rabbits without treatment (G2) (Fig. 6C) or treated with 3.5 mg/kg (G3) (Figs. 6D), 17 mg/kg of aqueous extract (G4) (Fig. 6E) and 5 mg/kg of chlorpropamide (G5) (Fig. 6F) presented a decrease in the number and size of pancreatic islets, compared to control group (G1). In addition, the G2, G3 and G4 groups frequently presented increased amount of blood in the vessels (hyperemia) (Fig. 6C and F). The exocrine pancreas showed similar histological characteristics in all groups (Fig. 6A–F).

Our results showed that there was a reduction in the size of the endocrine portion of the pancreas of the diabetic animals (Fig. 6). As a result of this reduction, there is a decrease in the production of insulin and, as a consequence, the impairment of fatty acid synthesis in the liver, a reduction in the activity of lipoprotein lipase, a reduction in the production of digestive enzymes (e.g. lipase, trypsin, chymotrypsin,

elastase, etc.) and increased uptake of triacylglycerols by adipose tissue.

3.2. Butanolic extract

The plasma glucose concentration varied over time ($F = 13.548$; $p < 0.001$) but did not vary with treatment ($F = 1.408$, $p = 0.26$) and interaction between treatment and time ($F = 1.634$; $p = 0.14$). Glucose values were higher at 21 days ($F = 12.578$; $p < 0.001$) when compared to 1 and 15 days, which in turn were not different from each other (Fig. 7). At 15 days of treatment, the treated groups showed similar reductions. However, at 21 days G3 showed reductions of 32.11% and G4 showed a percentage reduction of 37.8% when compared to the untreated diabetic group (G2) and a significant reduction when compared to G5 (Table 4). The lowering of glucose values can be attributed to a synergistic effect of flavonoids present in the extract according to several studies that point to glucose reduction in animals treated with flavonoids (Vessal et al., 2003; Nuraliev & avezov, 1992; Laddha et al., 2019).

The results showed that the concentration of triacylglycerol did not vary during treatment ($F = 2.544$, $p = 0.06$) or simply as a function of time ($F = 17.019$, $p < 0.001$) at 5% probability, but significant at 10%. In general, triglyceride concentration was highest after 21 and significantly greater than for 15 days of treatment ($F = 17.019$, $p < 0.001$) (Fig. 8).

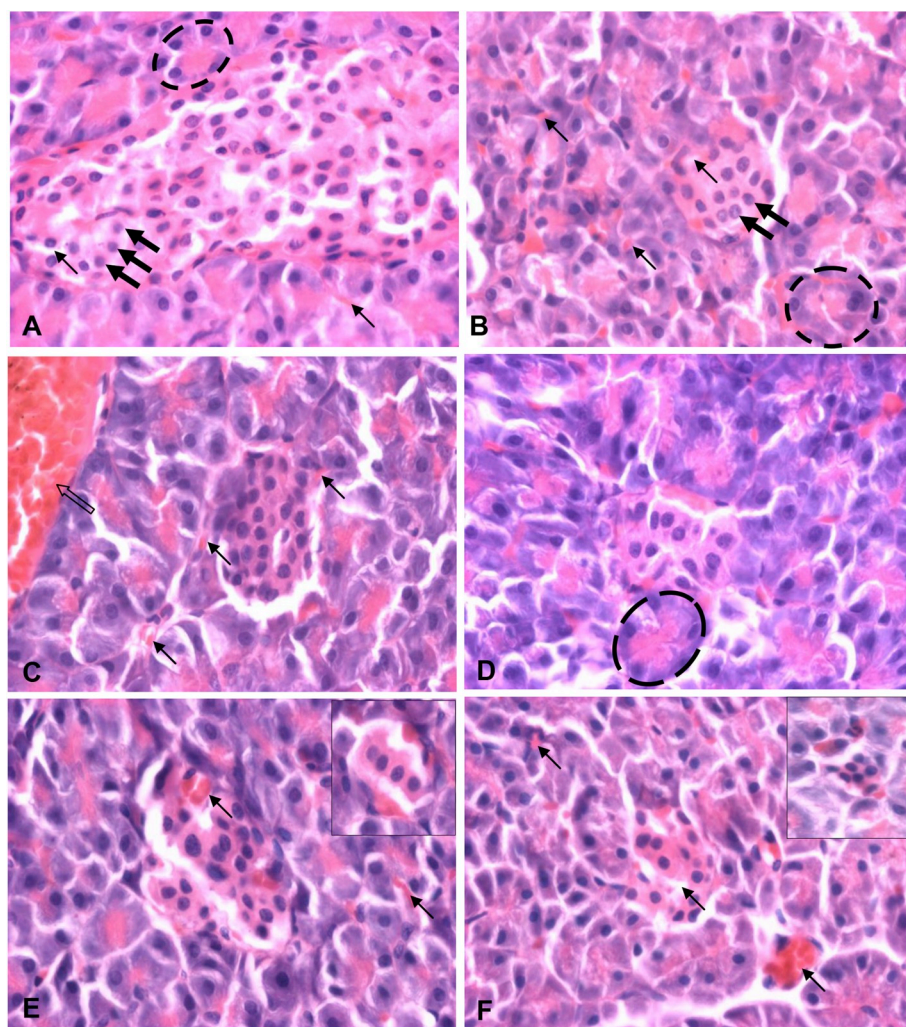


Fig. 6. Histological aspects of pancreas of control rabbits (A and B), untreated diabetics (C) and diabetics treated with 3.5 mg/kg of aqueous extract (D), 17 mg/kg of aqueous extract (E) and with Chlorpropamide at a dose of 5 mg/kg (F) after 23 days of treatment. Photomicrographs of pancreas sections stained with hematoxylin and eosin. 400X magnification. A and B: The control pancreas (G1) presents preserved parenchyma showing an exocrine portion formed by serous acini (circled) that surrounds an endocrine portion consisting mostly of large-diameter pancreatic islets, showing cellular cords (thick arrows). Observe small blood capillaries between the acini and inside the pancreatic islets (thin arrows). C: Alloxane-diabetic pancreas without treatment (G2) presenting few and small pancreatic islet. Note a hyperemic vessel (empty arrow) and small blood capillaries between the acini and inside the pancreatic islet (thin arrows). D: Alloxane-diabetic pancreas treated with 3.5 mg/kg of aqueous extract (G3) presenting a small pancreatic islet. Note pancreatic acini with the same characteristics of the control group (circled). E: Alloxan-diabetic pancreas treated with 17 mg/kg of aqueous extract (G4) showing small pancreatic islets containing few cells (in the center of image and in the insert). Small blood capillaries between the acini and inside the pancreatic islet (thin arrows). F: Alloxan-diabetic pancreas treated with 5 mg/kg of chlorpropamide (G5) showing small pancreatic islets containing few cells (in the center of image and in the insert). Small blood capillaries between the acini and inside the pancreatic islet (thin arrows). Hyperemic vessel (thin right arrow).

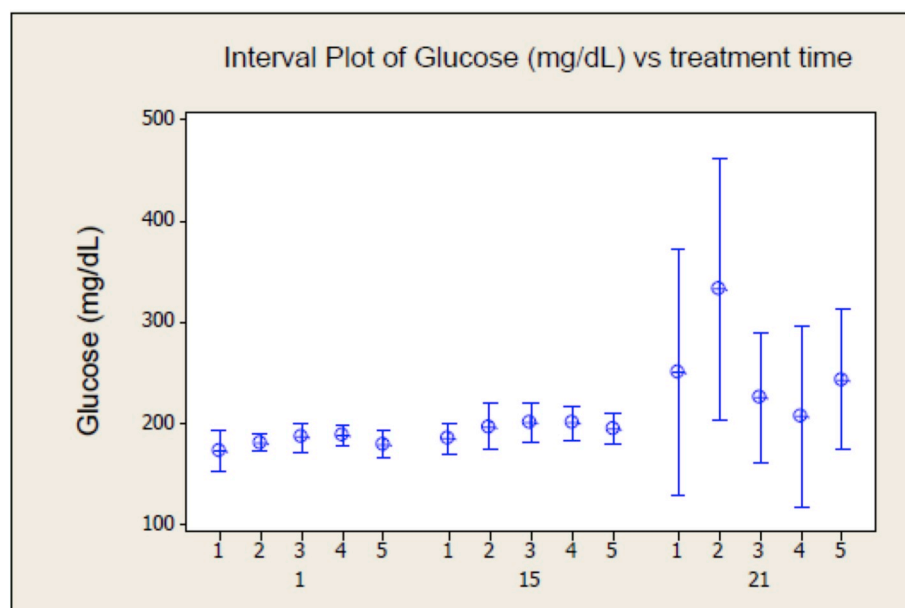


Fig. 7. Mean values and confidence levels of blood glucose levels for the control group (C), untreated diabetic rabbits (D), diabetic rabbits treated with butanolic extract at a concentration of 3.5 mg/kg (DEB 3.5), 17 mg/kg (DEB 17) and 5 mg/kg Chlorpropamide (DC) at 15 and 21 days.

Table 4

Percentage of glucose variation for all groups in relation to the diabetic group without treatment and the diabetic group treated with chlorpropamide after the 15th and 21st day of treatment.

Groups	GLUCOSE			Groups			
	1 Days	15 Days	21 Days		1 Days	15 Days	21 Days
G2				G5			
G1	−4.83%	−5.84%	−24.66%*	G1	−3.8%	−4.52%	+3.08%*
G3	+2.75%	+1.70%	−32.11%*	G3	+3.86%	+3.12%	−7.51%*
G4	+3.85%	+1.83%	−37.8%*	G4	−4.67%	+3.26%	−14.89%*
G5	−1.07%	−1.37%*	−26.91%*	G2	+1.08%	+1.39%	+36.83%*

Legend: The averages with * differ from the G2 standard and the G5 standard at the 5% probability level by the Dunnet test. G1: Normal group. G2: Diabetic group without treatment. G3: Diabetic group that received butanolic extract at a concentration of 3.5 mg/kg. G4: Diabetic group that received butanolic extract at a concentration of 17 mg/kg. G5: Diabetic group treated with chlorpropamide.

According to Table 5, the group treated with chlorpropamide (G5) showed a reduction of 25.94% in the levels of triacylglycerols after 15 days of treatment, 21 days there was a reduction of 32.70% for G3 and a reduction of 24.62% for G4, when compared to G2. These reductions may be directly associated with the effects of flavonoids in increasing lipase activity.

Flavonoid compounds have the ability to decrease blood lipid concentration (Nagem et al., 1994, 1995) or lipase activity include morine, naringenin, naringin and rutin (Lima et al., 1999; Gomes, 1998). Excess triacylglycerol is stored in the liver and adipose tissue. Hepatocytes in groups G1, G3, G4 and G5 showed a normal pattern. Hepatocytes are polyhedral, granular, acidophilic cytoplasm, having one or two loose chromatin nuclei. The hepatocytes of the G2 group showed macroscopic aspects typical of hydropic degeneration in the initial stage, ballooned form and its rarefied or pale cytoplasm. In addition, lipids accumulate (steatosis) at an early stage. Non-alcoholic steatohepatitis is directly related to diabetes mellitus. These results allow us to conclude that the two dosages of the butanolic extract were effective against the accumulation of fats and in reducing the several injuries in the liver tissue (Fig. 9).

The histological analysis of the alloxane-diabetic pancreas without treatment or treated with different concentrations of the butanolic extract of *Equisetum giganteum* and with the drug chlorpropamide is represented in Fig. 10. There was no histological difference between untreated diabetic animals (G2) and treated animals (G3, G4 and G5).

Fig. 11 shows the means and confidence intervals for plasma cholesterol levels. The G3 group presented the lowest cholesterol values and plasma cholesterol concentration varied significantly between the G3 group and the other groups ($F = 3.920$, $p = 0.013$). No difference was found between groups G1, G2, G4 and G5 ($LRT = 2.591$, $p = 0.12$).

Groups G3 and G5 showed the highest percentages of reduction, 29.39% and 37.06%, respectively at 21 days when each group was compared with G2, the diabetic group without treatment (Table 6).

Studies regarding the hypocholesterolemic effects of flavonoids in experimental models of diabetes showed that rabbits treated with flavonoid morina at a concentration of 5 mg/kg had lower blood glucose values (83.1 mg/kg) when compared to the diabetic group (135.73 mg/Kg) (Lopes et al., 2000a,b). In a study of hyperlipidemia in rabbits, the combination of naringenin (5 mg/kg) and chitosan (30 mg/kg) were found to be effective in controlling lipid metabolism. These studies suggest that the flavonoids present in the butanolic extract of *Equisetum giganteum* may act to inhibit HMG-CoA reductase.

The plasma albumin concentration varied between the treated group G4 (17 mg/kg of butanolic extract) and the other groups G1, G2, G3 and G5. Furthermore, G4 had the lowest albumin values and at 21 days presented the highest percentage of reduction (33.84%) when compared to the diabetic group without treatment (G2).

The plasma creatinine concentration varied with time ($F = 80.225$; $p < 0.001$) and interaction between treatment and time ($F = 2.799$; $p = 0.05$). It was observed that the treatments were not statistically

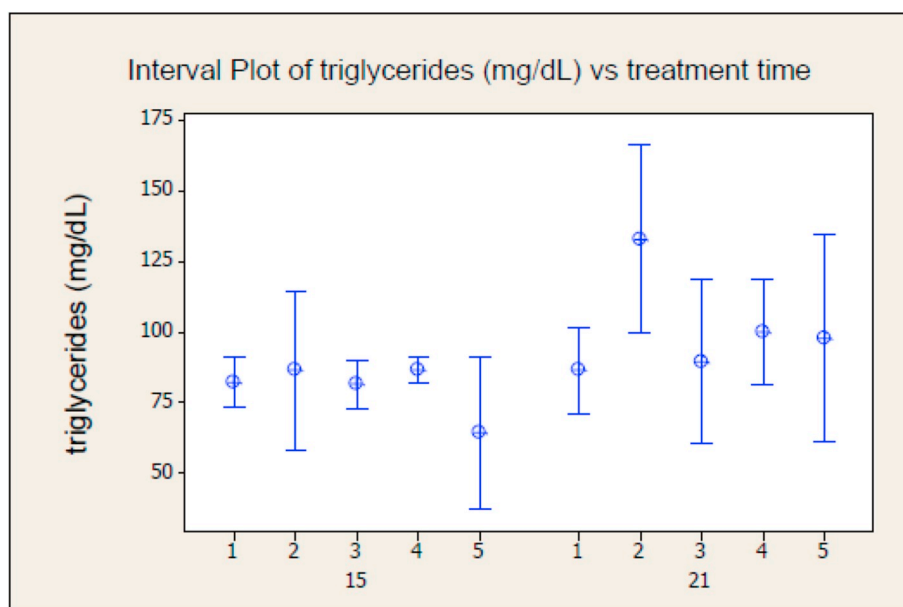


Fig. 8. Mean values and confidence levels of blood triglycerides levels for the control group (C), untreated diabetic rabbits (D), diabetic rabbits treated with butanolic extract at a concentration of 3.5 mg/kg (DEB 3.5), 17 mg/kg (DEB 17) and 5 mg/kg Chlorpropamide (DC) at 15 and 21 days.

Table 5

Percentage variation of the triacylglycerol for all groups in relation to the diabetic group without treatment and the diabetic group treated with chlorpropamide after the 15th and 21st day of treatment.

Triglycerides					
Groups	15 Days	21 Days	Groups	15 Days	21 Days
G2			G5		
G1	−4.81%	−35.03% ^a	G1	28.53%	−11.48%
G3	−5.92%	−32.70% ^a	G3	27.05%	−8.31%
G4	0.13%	−24.62% ^a	G4	35.21% ^a	2.70%
G5	−25.94%	−26.60% ^a	G2	35.04% ^a	36.24% ^a

^a Values differ from the G2 standard and the G5 standard at the 5% probability level (Dunnet test). G1: Normal group. G2: Diabetic group without treatment. G3: Diabetic group that received butanolic extract at a concentration of 3.5 mg/kg. G4: Diabetic group that received butanolic extract at a concentration of 17 mg/kg. G5: Diabetic group treated with chlorpropamide.

significant at the 5% probability level ($F = 1.157$; $p = 0.35$). The creatinine values were highest after 21 days ($F = 64.277$; $p < 0.001$) and the G5 group presented highest percentage increase (+17.04%) of creatinine when compared to the G2 group. At 21 days, the group treated with 3.5 mg/kg of extract (G3) reduced creatinine the most

(7.83%) when compared to G2 and G5 (5.51%). At the 15 day interval, G3 again showed a greater reduction rate of 16.51% when compared to G5. Based on these results, there seems to be no kidney damage in these animals. It is worth mentioning that the normal creatinine level for rabbits ranges from 0.8 to 2.57 mg/kg and the results observed in this study are within this range.

Regarding glycosylated hemoglobin, statistical analysis revealed that the effect of treatment with the butanolic extracts did not significantly alter blood glycosylated hemoglobin levels ($F = 2.604$; $p = 0.63$) after 21 days (Fig. 12).

The group treated with 3.5 mg/kg of butanolic extract (G3) reduced glycosylated hemoglobin by 14.54% (Table 7) and when comparing with chlorpropamide (G5), group G3 showed the highest percentage of reduction (13.55%).

Treatment with the butanolic extract caused statistically significant alterations at the 5% probability level ($F = 14.301$; $p = 0.006$) for the lipase enzyme. The untreated diabetic group (G2) had lower lipase activity than all the other groups (Fig. 13).

The group treated with 17 mg/kg of the butanolic extract (G4) showed greater lipase activity (Table 8) and in this case, an increase of 143.92% at 21 days when compared to the diabetic group without treatment (G2) was observed. The G3 group increased lipase activity by 114.55%. These results demonstrated that the hypolipidemic effect seen

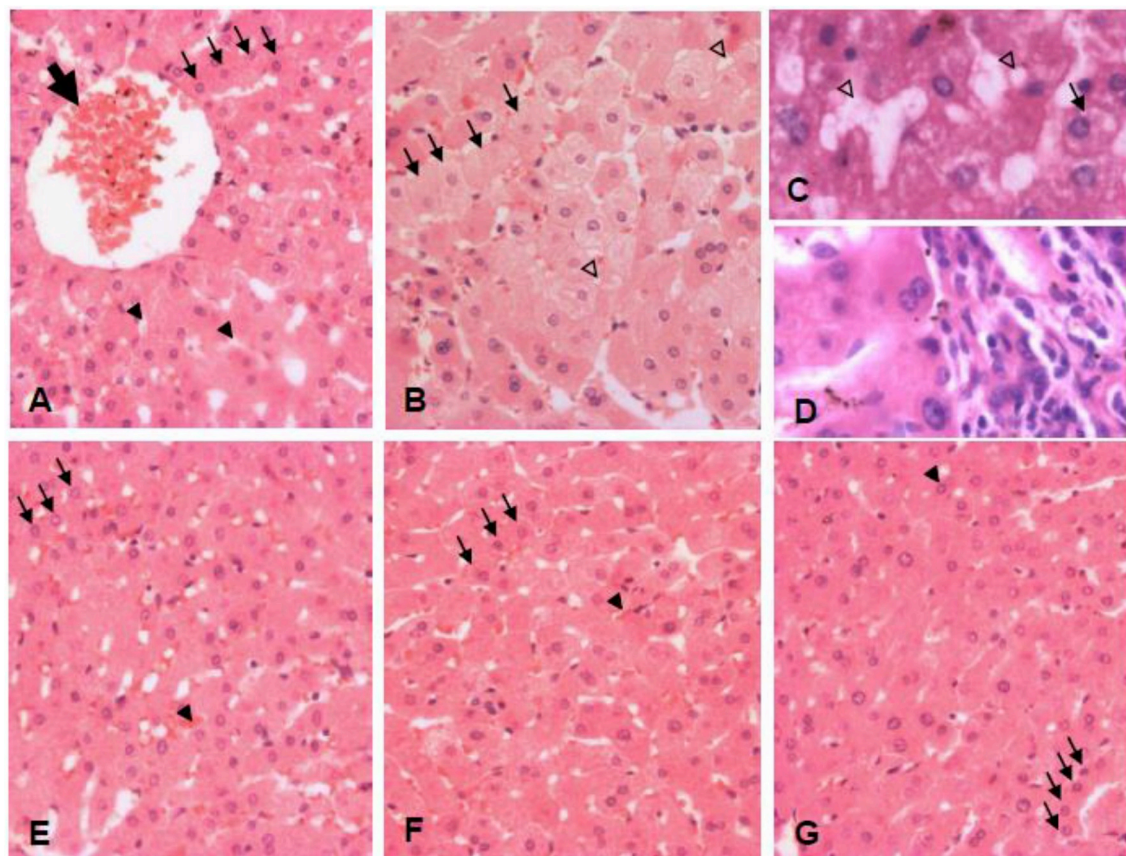


Fig. 9. Histopathological aspects of the liver. Photomicrographs of liver sections stained with hematoxylin and eosin. 200X magnification (A, B, E, F, G) and 400X magnification (B, C). Representative image of the liver of control rabbits (A), untreated diabetics (B, C, D) and diabetics treated with 3.5 mg/kg of butanolic extract (E), with 17 mg/kg of butanolic extract (F) and with 5 mg/kg of chlorpropamide (G) after 21 days of treatment. A: Control liver (G1) presents preserved liver parenchyma showing central vein (thick arrow) and hepatocyte plaques (thin arrows) limiting spaces occupied by well-preserved sinusoid capillaries (arrowheads). B: Alloxane-diabetic liver without treatment (G2) showing hepatocyte plaques (thin arrows) containing enlarged cells with morphological aspect of steatosis (thin arrow) and sinusoidal dilatation (empty arrowheads), more frequent in these animals when compared to the other groups. C: Detail of alloxane-diabetic liver hepatocytes without treatment (G2) that presents large hepatocytes with morphological aspect of steatosis (arrow) and sinusoidal dilatation (empty arrowheads). D: Detail of the periportal region of the alloxane-diabetic liver without treatment (G2), containing an inflammatory process with a predominance of mononuclear cells (right region). E, F and G: Alloxane-diabetic liver treated with 3.5 mg/kg of butanolic extract G3 (E), with 17 mg/kg of butanolic extract G4 (F) and with 5 mg/kg of chlorpropamide G5 (G) presenting histological characteristics similar to those of the control group. Hepatocyte plates (thin arrows), well-preserved sinusoid capillaries (arrowheads).

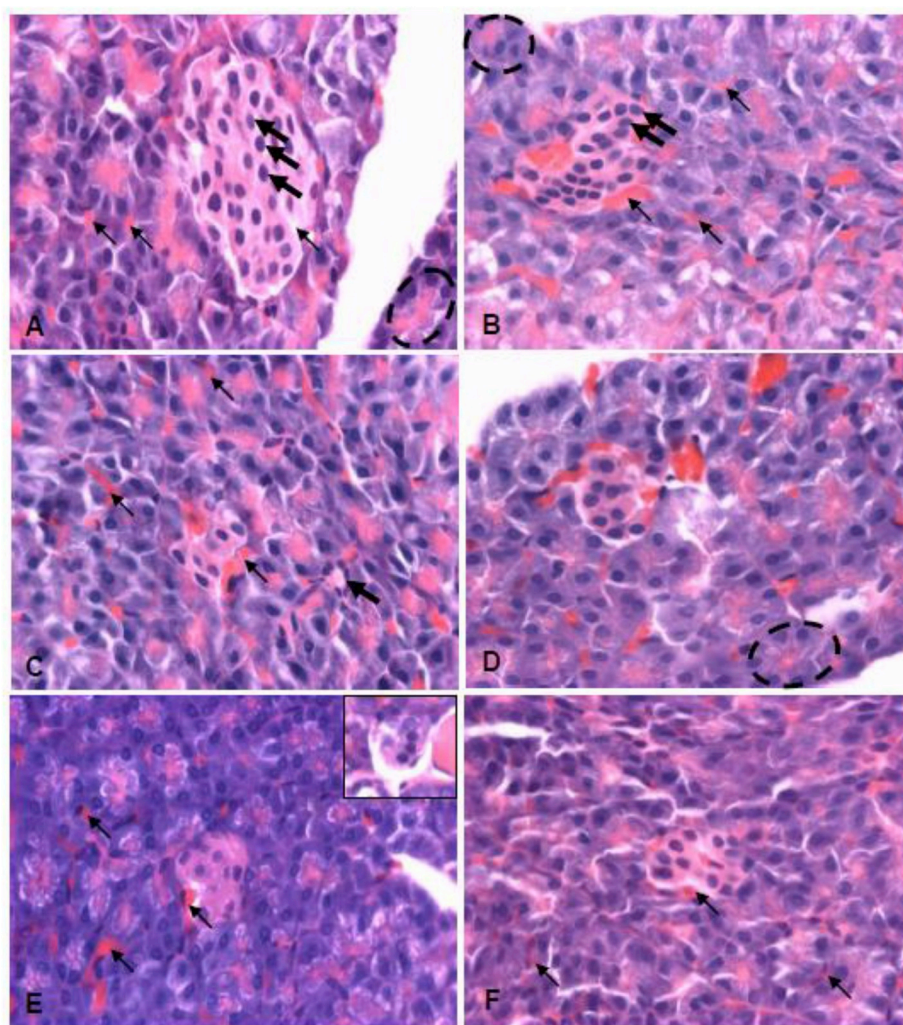


Fig. 10. Histological analysis of rabbit pancreas from the control groups (A and B), untreated diabetic rabbits (C), and diabetics rabbits treated with 3.5 mg/kg butanolic extract (D), 17 mg/kg butanolic extract (E) and 5 mg/kg Chlorpropamide (F) after 23 days of treatment. Photomicrographs of pancreas sections stained with hematoxylin and eosin. 400X magnification. **A and B:** The control pancreas presents preserved parenchyma showing an exocrine portion formed by serous acini (circled) that surrounds an endocrine portion consisting mostly of large-diameter pancreatic islets, showing cellular cords (thick arrows). Small blood capillaries between acini and the inside of the pancreatic islets are noted (thin arrows). **C:** The pancreas of alloxan induced diabetic rabbits that were not treated present small pancreatic islet that contain few cells. It is noteworthy that here there are small blood capillaries between acini and inside of the pancreatic islet (thin arrows) and intercalated duct (bold arrow) component of the exocrine pancreas. **D:** Alloxan-diabetic pancreas treated with 3.5 mg/kg of butanolic extract (DEB 3.5) displayed few and small pancreatic islets (in the center of the image and in the insert). The pancreatic acini have the same characteristics as the control pancreas (circled). **E:** The pancreas of alloxan induced diabetic rabbits treated with 17 mg/kg butanolic extract (DEB 17) showing small pancreatic islets containing few cells (in the center of image and in the insert). Small blood capillaries between the acini and within the pancreatic islet (thin arrows). **F:** The pancreas of alloxan induced diabetic rabbits treated with 5 mg/kg Chlorpropamide (DC) showing small pancreatic islets containing few cells (in the center of image and in the insert). Small blood capillaries between the acini and within the pancreatic islet (thin arrows).

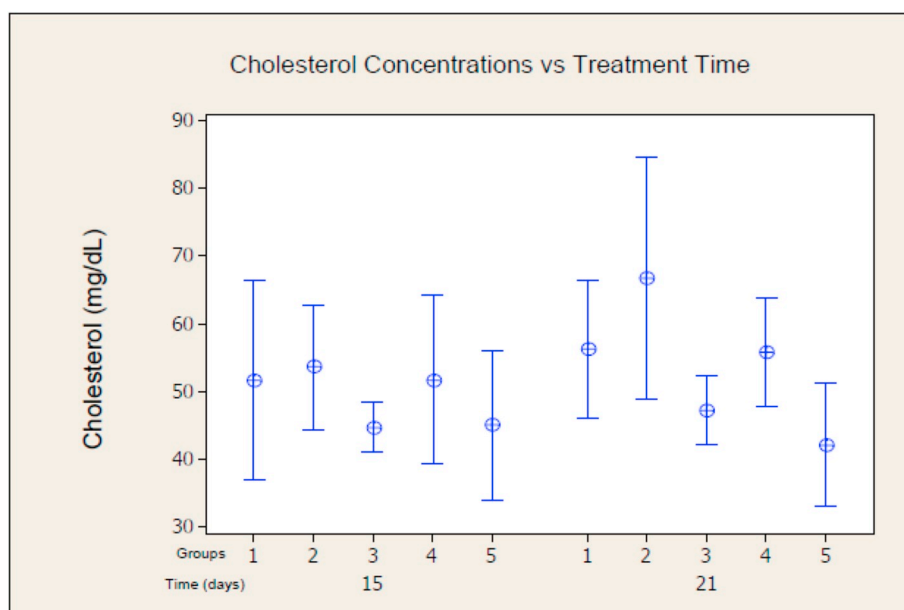


Fig. 11. Means and confidence intervals of plasma cholesterol levels for control animals (G1), diabetics without treatment (G2), diabetics treated with butanolic extract at a concentration of 3.5 mg/kg (G3), 17 mg/kg (G4) and 5 mg/kg of chlorpropamide (G5) at 15 and 21 days.

Table 6

Percentage of cholesterol variation for all groups in relation to the diabetic group without treatment and the diabetic group treated with chlorpropamide after the 15th and 21st day of treatment.

Groups	Cholesterol		Groups		
	15 Days	21 Days		15 Days	21 Days
G2			G5		
G1	−3.53%	−15.73%	G1	+14.99%	+33.89% ^a
G3	−16.57%	−29.39% ^a	G3	−0.55%	+12.24%
G4	−3.53%	−16.34%	G4	+14.99%	+32.91% ^a
G5	−16.10% ^a	−37.06% ^a	G2	+19.19% ^a	+58.88% ^a

^a Values differ from the G2 standard and the G5 standard at the 5% probability level (Dunnet test). G1: Normal group. G2: Diabetic group without treatment. G3: Diabetic group that received butanolic extract at a concentration of 3.5 mg/kg. G4: Diabetic group that received butanolic extract at a concentration of 17 mg/kg. G5: Diabetic group treated with chlorpropamide.

in this study is due, in part, to the increase in lipase activity in the treated and normal groups. When comparing each group with the G5 group (diabetic treated with chlorpropamide), it was found that G2 (diabetic without treatment) reduced lipase activity the most by 55.25%.

The butanolic extract had a significant effect on this parameter, causing an increase in lipase activity, which in the case of diabetes is important given that hypertriglyceridemia is present in diabetes and it is therefore useful to increase the activity of the enzyme that hydrolyzes triacylglycerols.

3.3. Phytochemical study

Analysis of the chromatograms and UV spectra of the butanolic extract revealed the presence of eight classes of flavonoids: flavones, flavonols, dihydroflavones, flavanones, chalcones, aurones and iso-flavones (Fig. 14).

Although the aqueous extract is less rich in flavonoids than the butanolic extract, it similarly contained commonly encountered flavonoid compounds (Fig. 15).

The UV spectra for the principal peaks in both chromatograms are consistent with the spectroscopic profile of flavonoid compounds

Table 7

Percentage variation of glycosylated hemoglobin for all groups in relation to the diabetic group without treatment and the diabetic group treated with chlorpropamide after the 15th and 21st day of treatment.

Groups	Glycosylated Hemoglobin		Groups		
	21 Days			21 Days	
G2			G5		
G1	−13.78%		G1	−12.79%	
G3	−14.54% ^a		G3	−13.55% ^a	
G4	−9.48%		G4	−8.44%	
G5	−1.13%		G2	+1.15%	

^a Values differ from the G2 standard and the G5 standard at the 5% probability level (Dunnet test). G1: Normal group. G2: Diabetic group without treatment. G3: Diabetic group that received butanolic extract at a concentration of 3.5 mg/kg. G4: Diabetic group that received butanolic extract at a concentration of 17 mg/kg. G5: Diabetic group treated with chlorpropamide.

described in the literature (Marcucci et al., 1998). Flavonoids have two absorption maxima: one occurring between 240 and 285 nm (band II) related to the benzenoid ring and another between 300 and 400 nm (band I) due the cinnamic acid functionality. Flavones absorb at 310–350 nm, flavonols (3-O-substituted) of 330–360 nm and flavonols at 350–385 nm (Markham, 1982). An increase in oxygen atoms on the B-ring of flavones and flavonols results in a bathochromic effect on band I, whilst maintaining the same value for band II. Band II may have one or two peaks (designated IIa and IIb) depending on the oxygenation pattern on the ring B. The presence of hydroxyl groups in flavones and flavonols gives rise to a significant bathochromic effect on band II and a smaller effect on band I (Mabry et al., 1970).

Given the fact that phenolic compounds may be bound to other elements such as carbohydrates and proteins (Mattila et al., 2000), these linkages were cleaved by acidic hydrolysis to release any bound flavonoids prior to mass spectrometry analysis. The phenolic composition of *E. giganteum* stems was studied by liquid chromatography coupled to diode array detection (LC–DAD) and liquid chromatography coupled to electrospray ionization-tandem mass spectrometry (LC–ESI–MS/MS) by Francescato and co-workers in 2013 (Francescato et al., 2013). They identified caffeic acid derivatives, flavonoids, styrylpyrones and the most abundant glycosylated flavonoids in this sample were

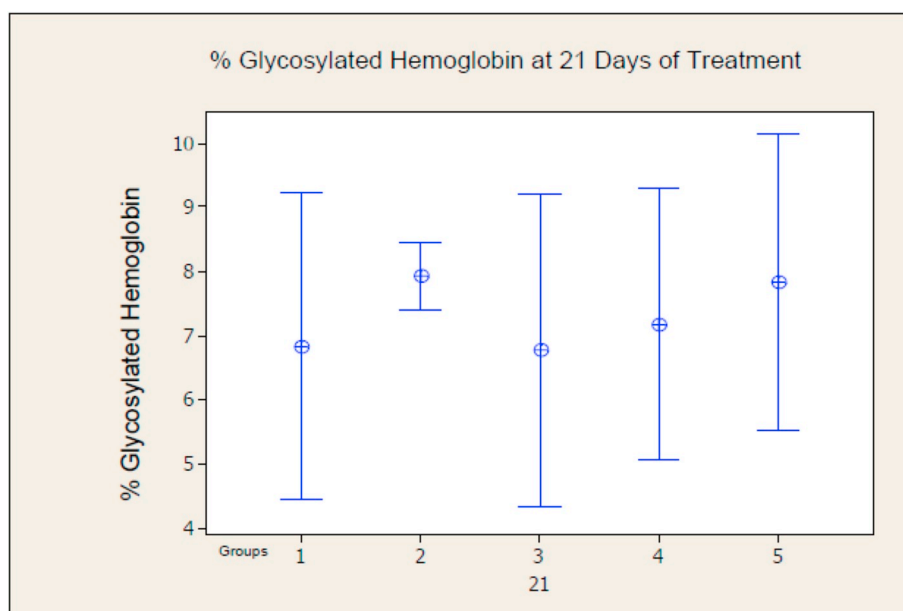


Fig. 12. Means and confidence intervals of glycosylated hemoglobin (HG) blood levels for control animals (G1), diabetics without treatment (G2), diabetics treated with butanolic extract at a concentration of 3.5 mg/kg (G3), 17 mg/kg (G4) and 5 mg/kg of chlorpropamide (G5) at 21 days.

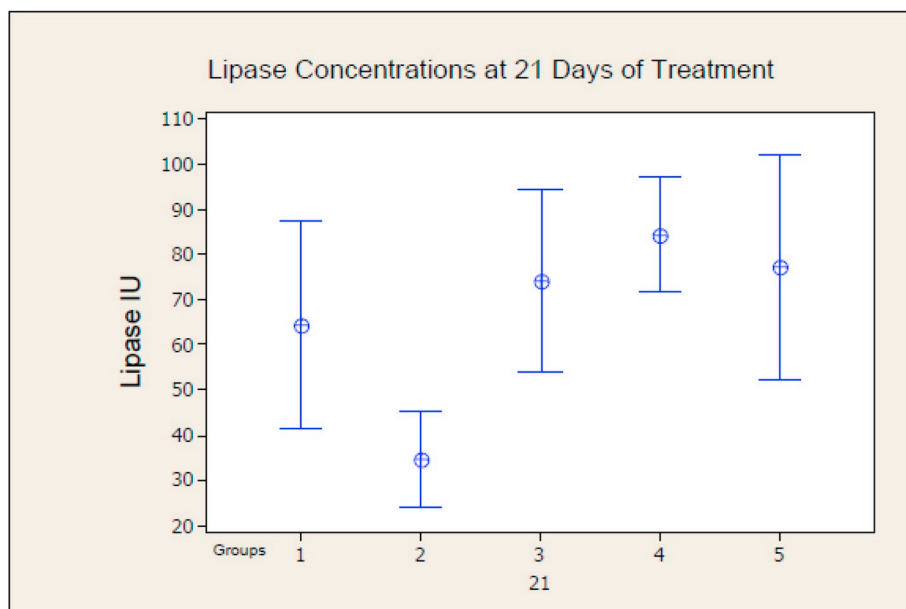


Fig. 13. Means and confidence intervals of blood lipase levels for control animals (G1), diabetics without treatment (G2), diabetics treated with butanolic extract at a concentration of 3.5 mg/kg (G3), 17 mg/kg (G4) and 5 mg/kg of chlorpropamide (G5) at 21 days.

Table 8

Percentage of variation of lipase for untreated diabetic and diabetic animals treated with Diabinese after the 21st day of treatment.

TABLE 1. Mean cholesterol levels on the last day of treatment.				
Groups	Lipase		Groups	21 Days
	21 Days			
G2			G5	
G1	+ 86.40%		G1	− 16.58% ^a
G3	+ 114.55% ^a		G3	− 3.98%
G4	+ 143.92% ^a		G4	+ 9.17% ^a
G5	+ 123.44% ^a		G2	− 55.25% ^a

^a Values differ from the G2 standard and the G5 standard at the 5% probability level (Dunnet test). G1: Normal group. G2: Diabetic group without treatment. G3: Diabetic group that received butanolic extract at a concentration of 3.5 mg/kg. G4: Diabetic group that received butanolic extract at a concentration of 17 mg/kg. G5: Diabetic group treated with chlorpropamide.

kaempferol derivatives. In addition, quercetin-3-O-(caffeoyl)-glucoside and 3-hydroxyhispidin-3,4'-di-O-glucoside, were reported for first time in the *Equisetum* genus.

The LC-UV/VIS chromatographic profiles of both the butanolic and

aqueous extracts of *E. giganteum* extract are shown in Figs. 16 and 17.

The molecular mass of the compounds was obtained from their positive ion electrospray mass spectra (ESI-MS), which described the corresponding protonated pseudomolecular ions as well as some doubly-protonated adduct ions (Table 9).

In accordance with the study described Francescato (Francescato et al., 2013) flavonoid glycoside quercetin-3,7-di-O-glucoside and equisetumprone were both observed in the butanolic extract. In the case of the aqueous extract, the main peak eluting at 14.23 min was tentatively identified by analysis of its mass spectrum to correspond to a ferulic acid ester derivative. Both Galangin and Naringenin co-eluted given the similarity in their chemical structures and higher molecular weight flavonoid glycosides eluted later in the HPLC run. Other known natural products such as Pinocembrin, Jaceosidin, Apigenin, Naringin, Quercetin and Kaempferol were also detected in the *E. giganteum* extracts. Kaempferol has been observed to reduce the relative risk of type 2 diabetes in people with an increased intake of the kaempferol containing foods (Song et al., 2005; López-Lázaro et al., 2011). Our studies converge with information in the literature regarding the various pharmacological and biological properties of flavonoids. As an example, among other identified flavonoids (Table 9), the Kaempferol is a

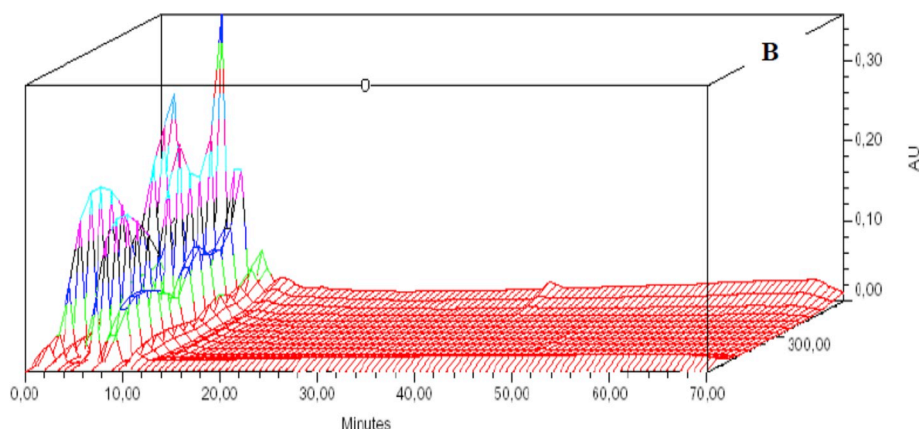


Fig. 14. Chromatogram and UV spectra in 3D for butanolic extract analyzed by HPLC. Conditions: ODS column (250 × 4.0 mm I.D., 5 μm), flow rate 1.0 ml/min, injection volume 10 μL and diode array detector.

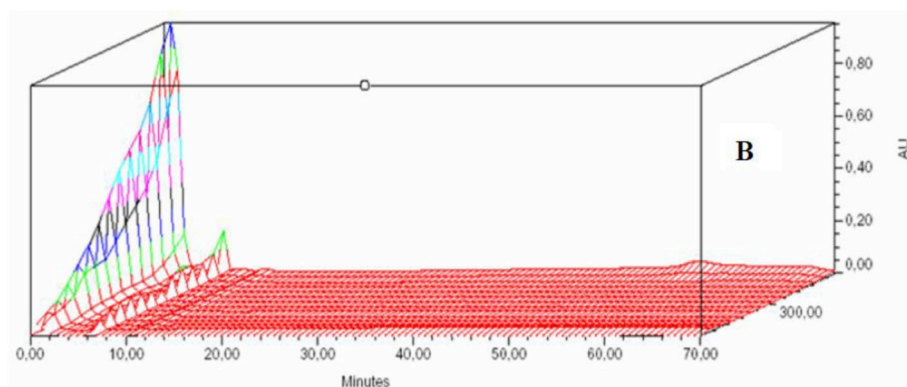


Fig. 15. Chromatogram and UV spectra in 3D for the aqueous extract analyzed by HPLC. Conditions: Eluent H₂O (A) and CH₃CN (B): 0 min 95% A, 5% B, 5 min 60% A, 95% B. ODS column (250 × 4.0 mm I.D., 5 μm), flow rate 1.0 ml/min, injection volume 10 μL and diode array detector.

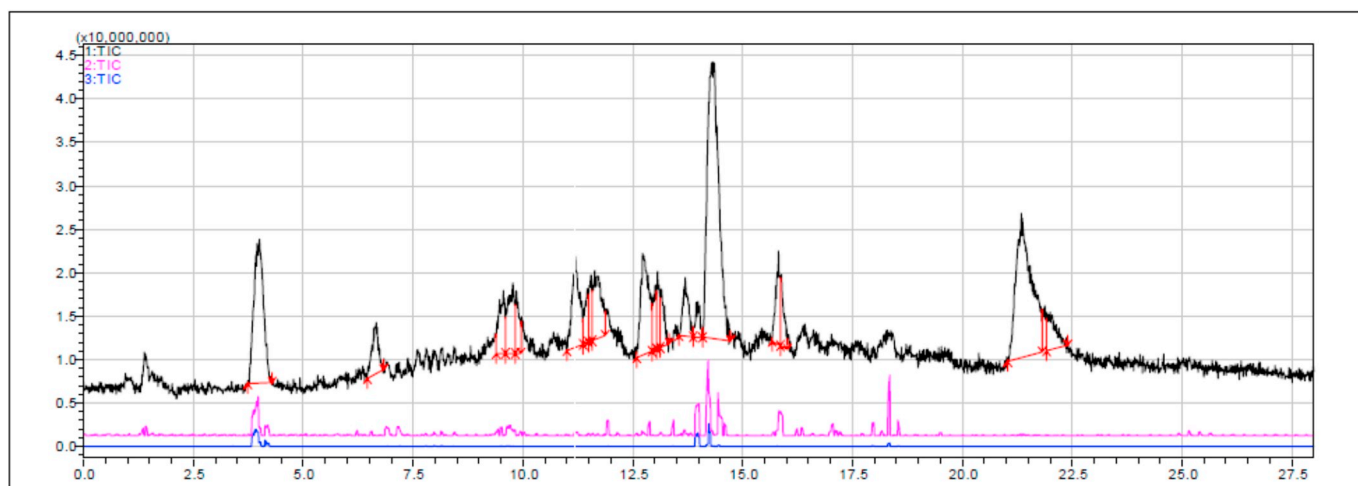


Fig. 16. HPLC-MS analysis of aqueous extract of *Equisetum giganteum*. Conditions: VP-ODS column (150 × 2.0 mm ID., 3 μm), flow 0.2 ml/min, temperature 40 °C, injected volume 5 μL, m/z range analyzed at 150 to 1000 u.m.a. Eluents: H₂O (A) and MeOH (B). Mobile phase gradient: 10% B to 70% B 15 min, 70% B isocratic for 10 min, 70–100% B for 2 min, 100 to 10% B in 3 min.

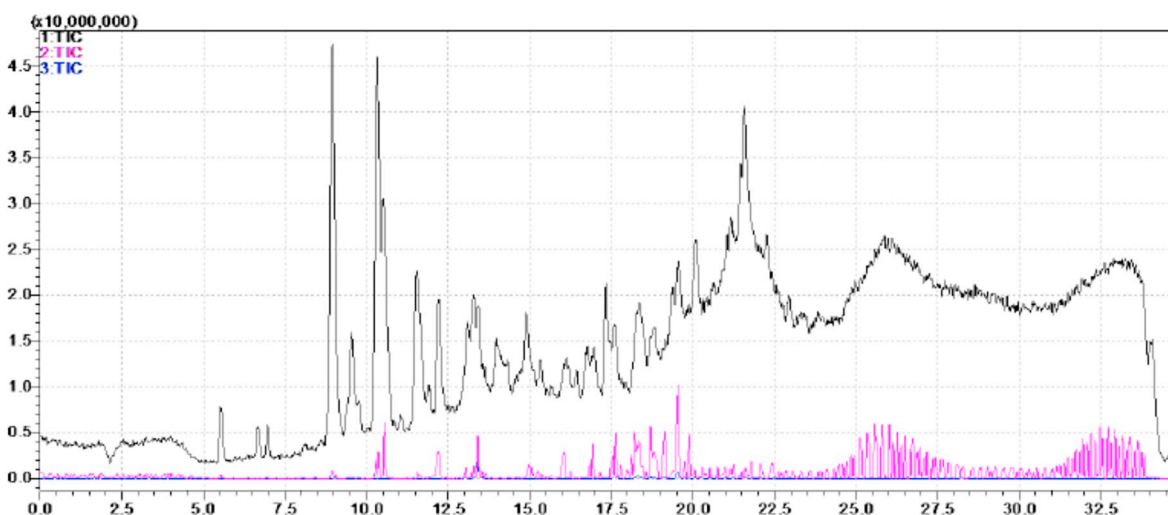


Fig. 17. HPLC-MS analysis of butanolic extract of *Equisetum giganteum*. Conditions: VP-ODS column (150 × 2.0 mm ID., 3 μm), flow 0.2 ml/min, temperature 40 °C, injected volume 5 μL, m/z range analyzed at 150 to 1000 u.m.a. Eluents: H₂O (A) and MeOH (B). Mobile phase gradient: 10% B to 70% B 15 min, 70% B isocratic for 10 min, 70–100% B for 2 min, 100 to 10% B within 3 min.

flavonoid widely is found in many edible plants (e.g. tea, broccoli, cabbage, kale, beans, endive, leek, tomato, strawberries and grapes) and in plants or botanical products commonly used in traditional

medicine (e.g. *Ginkgo biloba*, *Tilia* spp, *Equisetum* spp, *Moringa oleifera*, *Sophora japonica* and própolis). It has been estimated that the human dietary intake of this polyphenol may be up to approximately 10 mg/

Table 9
HPLC–MS data of phenolic compounds presents in *E. giganteum* extract.

Extract	R _t (min)	UV λ _{max} (nm)	[M + H] ⁺	Compound
Aqueous	3.83	275	182	Caffeic acid derivatives
Aqueous	6.49	265 and 346	257	Pinocembrin
Aqueous	12.62	275	331	Jaceosidin
Aqueous	14.23	265	311	Ferulic acid derivatives
Butanolic	6.51	265 and 346	303	Quercetin
Butanolic	6.89	255	271 and 273	Galangin and Naringenin
Butanolic	8.95	265 and 347	287	Kaempferol
Butanolic	11.26	266	425	Equisetum pyrone
Butanolic	13.85	287	271	Apigenin
Butanolic	17.92	287	581	Naringin
Butanolic	18.10	294	465	Quercetin-3-O-glucoside
Butanolic	20.62	318	627	Quercetin-3,7-di-O-glucoside

day. Epidemiological studies have found a positive association between the consumption of kaempferol containing foods and a reduced risk of developing cardiovascular diseases and some types of cancer. Numerous *in vitro* and some animal studies support a role of kaempferol in the prevention and/or treatment of these and other diseases, such as neurodegenerative diseases, infectious diseases, diabetes, osteoporosis, anxiety, allergies, inflammation and pain (López-Lázaro et al., 2011; De Vries et al., 1997; Jin-Ming et al., 2019).

4. Conclusion

The evaluation of the biochemical parameters for the aqueous extract of *E. giganteum* revealed that the aqueous extract reduced blood glucose and displayed high hypoglycemic effect. In the study of lipemia, cholesterol was reduced by 38% after 16 days and triacylglycerol was reduced by 50% over the same time period by the aqueous extract. Similar results were also observed for the butanolic extracts capacity to reduce glucose, cholesterol and triacylglycerol over a 16 or 21-day treatment period. Lipase activity was increased by 114% and glycosylated hemoglobin was reduced by 14% in animals treated with butanolic extract at 3.5 mg/kg and 9% at 17 mg/kg. Thus, treatment with aqueous and butanolic extract at dose of 17 mg/kg demonstrated both hypoglycemic and hypolipidemic effects, however the results of the aqueous extract were more expressive. The butanolic extract was more effective in reducing glycosylated hemoglobin and in significantly increasing lipase activity. It is important to note the positive hypoglycemic and hypolipidemic action of the extracts at concentrations up to 3.5 mg/kg and 17 mg/kg when compared to the group treated with chlorpropamide. The low concentration at which the extracts display biological activity is relevant when considering that the maximum daily dose of sulfonylurea used is 500 mg daily. There are several side effects observed among patients who, after five years of using sulfonylurea, about 50% of diabetics may have secondary failure (Araújo et al., 2000). The detection of different flavonoid classes in aqueous and butanolic extracts by HPLC-DAD and HPLC-MS suggest a synergism between the flavonoids present in the extracts that are responsible for the antidiabetic and antilipidemic properties. The histological analysis verified the absence of steatosis, of inflammatory infiltrate and hyperemia in the liver of the groups treated with aqueous and butanolic extracts. Therefore, the extracts were effective against fat accumulation and other damages in liver tissue, which represents complications of poorly controlled diabetes. Therefore, based on reductions in biochemical parameters, evidence of the absence of cytohistological complications in the liver and pancreas, these results suggest that *Equisetum giganteum* extracts can be a therapeutic resource in the treatment of diabetes and hyperlipidemia.

Author contributions

Geisla Teles Vieira (geislateles@yahoo.com.br) collected and identified of new plant occurrence sites the plant material, carried out in vivo experiments and phytochemical analysis as part of her Masters Degree studies (CIPHARMA/UFOP). Tânia Toledo de Oliveira (ttoledo@ufv.br) coordinated in vivo experiments and analyzed the experimental data. Gabriel Almeida Paes Humberto (gabriel.q.paes@gmail.com) prepared the plant extracts and samples for phytochemical characterization. Marco Antonio Alves Carneiro (marco.carneiro@ufop.edu.br) carried out statistical analysis on the data. Silvia Dantas Cangussu (cangussu@ufop.edu.br) studied the microscopic anatomy of biological tissues. Jason Guy Taylor (jason@iceb.ufop.br) analyzed the spectra and co-wrote the manuscript. Jorge Luiz Humberto (jorgedeq@ufop.edu.br) conceived and coordinated the research project and analyzed the spectra. All authors have read and approved the final manuscript.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2020.112898>.

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